基于神经网络的富语境关系自动抽取：一种新型的单语句编码器

基科42 唐路明

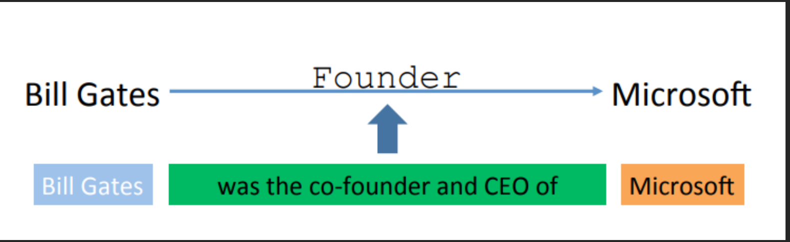
指导教师：刘知远 副教授

CONTENT

Part 1 知识图谱与关系抽取

知识图谱与关系抽取

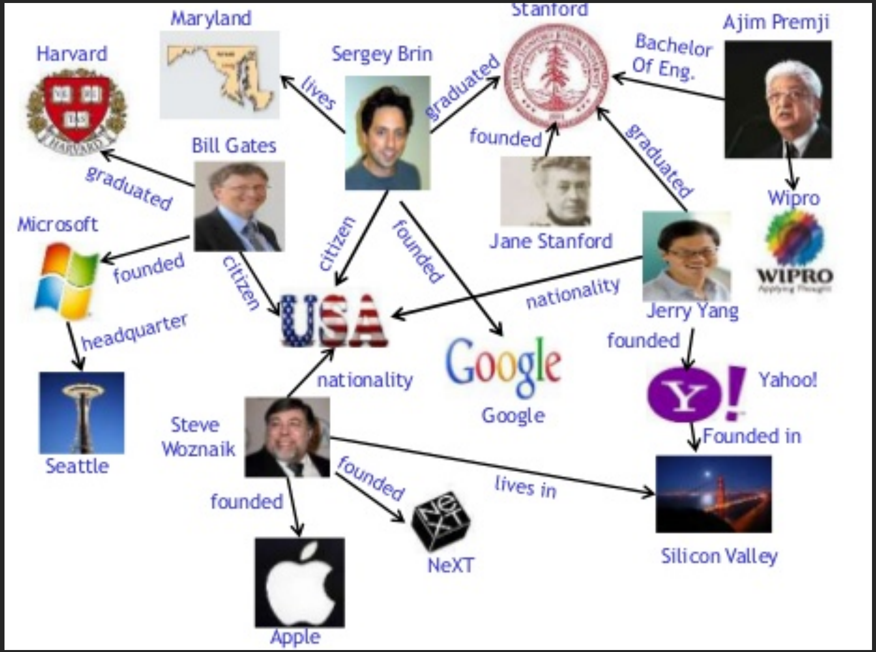
（头部实体， 关系类型， 尾部实体）



William Henry Gates III (born October 28, 1955) is an American business magnate, investor, author, philanthropist, and co-founder of the Microsoft Corporation along with Paul Allen.[2][3]

In 1975, Gates and Allen launched Microsoft, which became the world's largest PC software company. During his career at Microsoft, Gates held the positions of chairman, CEO and chief software architect, while also being the largest individual shareholder until May 2014.[4][a] Gates stepped down as chief executive officer of Microsoft in January 2000, but he remained as chairman and created the position of chief software architect for himself.[7]

原始文本



知识图谱

- 基于人工特征的关系抽取
  - 获取训练数据难度大
  - 语义学特征表示较差
- 远程监督
- 基于神经网络的关系抽取

- CNN：卷积核的短程滑动窗口难以捕捉长程依赖
- LSTM：梯度消散造成训练困难，时间长效率低
- 思考：是否有别的单语句向量表示方法？

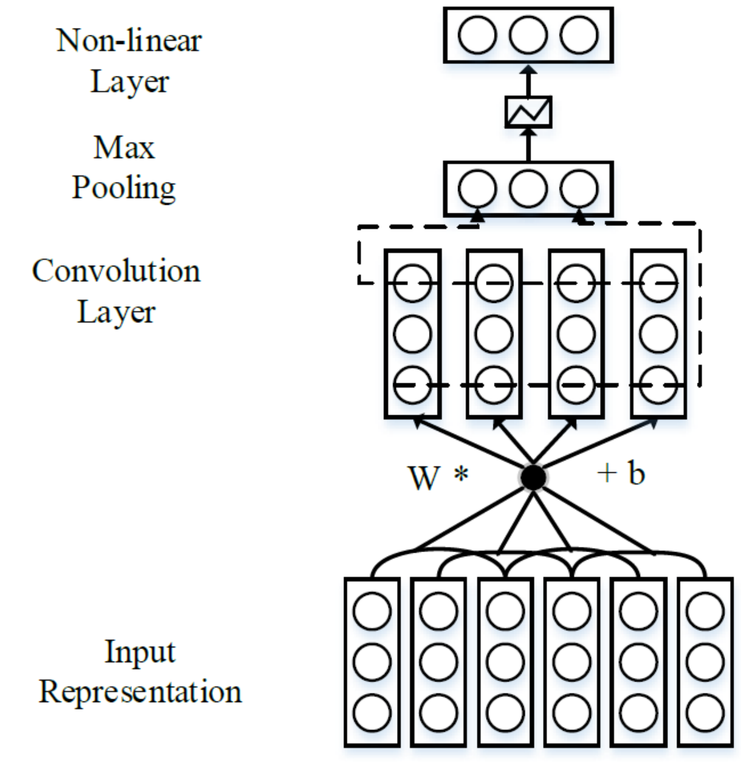
论文贡献

- 自主实现数个主流模型并在统一数据集上进行评测，建立较为统一的标准基线模型集合。
- 提出新型单语句编码器RE_NetVLAD，取得最好的单模型效果，不同设定下在效率和效果上都有稳定表现。
- RE_NetVLAD对于语句长度的变化鲁棒性更强。
- RE_NetVLAD模型内操作具有一定的可解释性的语言学意义。

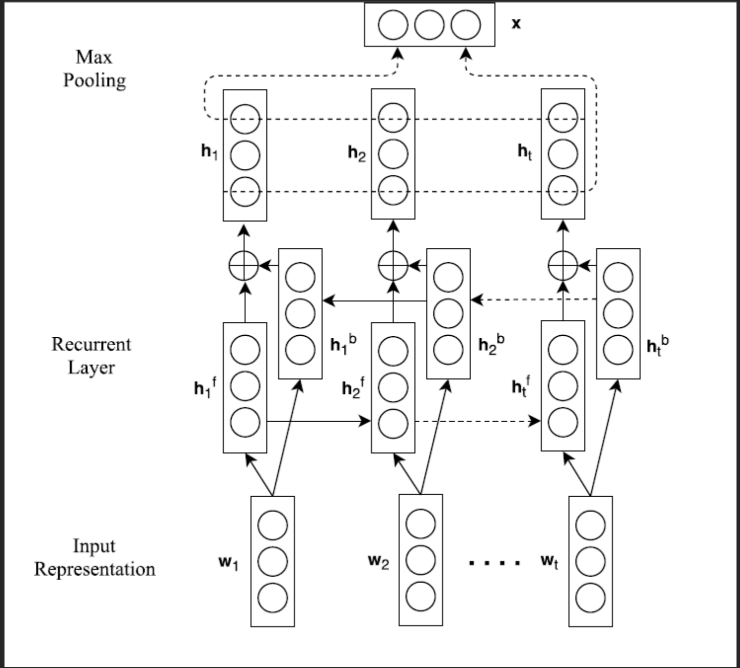
Part 2 神经关系抽取系统

- 输入：目标实体对+包含目标实体对的若干句话
  输出：目标实体对间的关系类型
- 结构：输入编码器+单语句编码器+多语句编码器
- 输入编码器：词嵌入向量+位置嵌入向量

单语句编码器

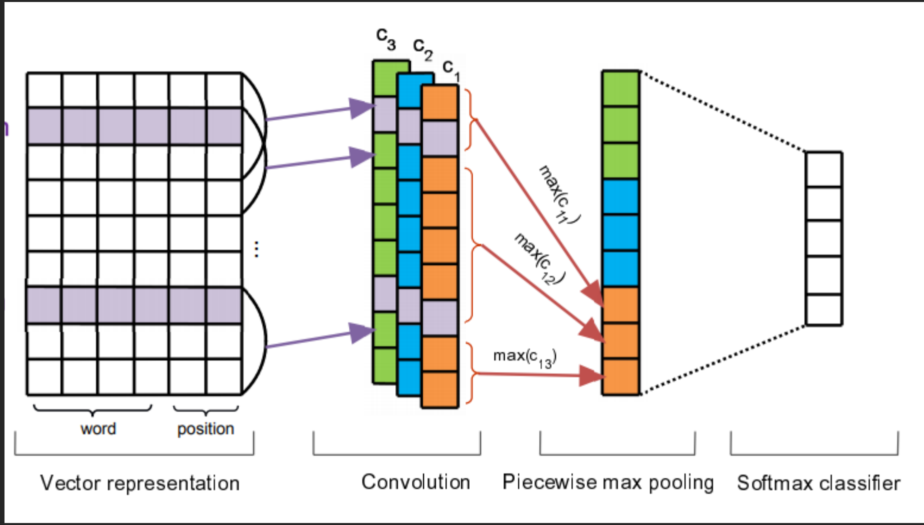


CNN

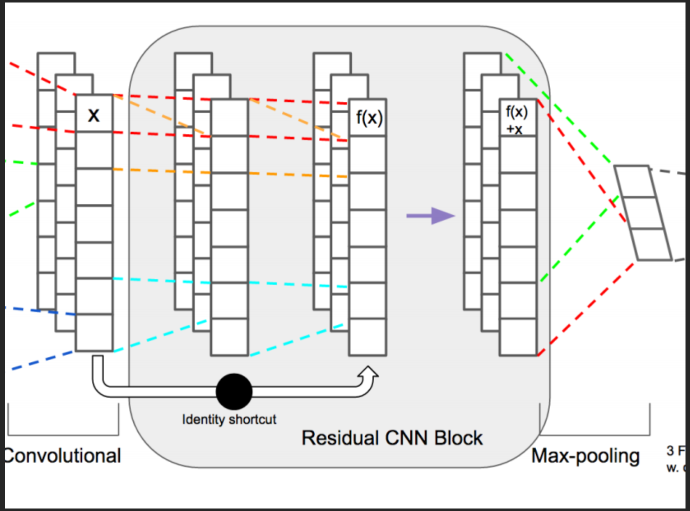


LSTM

单语句编码器



PCNN



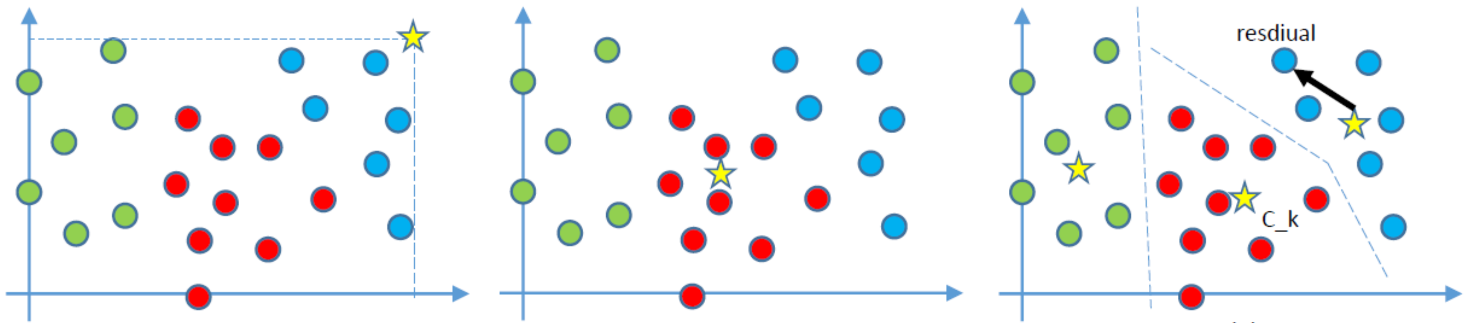
ResCNN

- 随机编码器：Rand
- 最大编码器：Max
- 平均编码器：AVG
- 基于注意力机制的编码器：ATT

Part 3 RE_NetVLAD

传统计算机视觉中的VLAD

$$V(j,k) = \sum_{i=1}^{N} a_k(\mathbf{x}_i)(x_i(j) - c_k(j)),$$



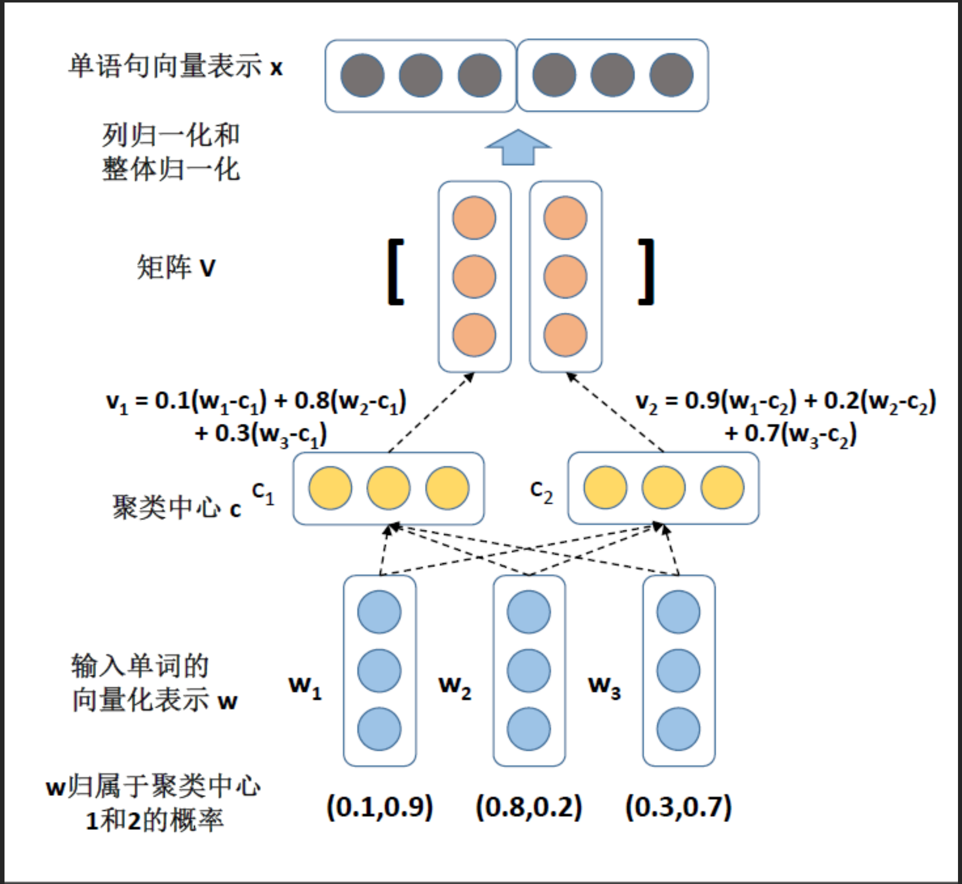
(a) (b) (c)

基于神经网络的NetVLAD

$$a_k(\mathbf{x}_i) = \frac{e^{\mathbf{w}_k^T \mathbf{x}_i + b_k}}{\sum_{k'} e^{\mathbf{w}_{k'}^T \mathbf{x}_i + b_{k'}}}$$

$$V(j,k) = \sum_{i=1}^{N} \frac{e^{\mathbf{w}_k^T \mathbf{x}_i + b_k}}{\sum_{k'} e^{\mathbf{w}_{k'}^T \mathbf{x}_i + b_{k'}}} (x_i(j) - c_k(j))$$

面向关系抽取的RE_NetVLAD



Part 4 实验结果与分析

数据集

- 数据集：NYT语料与Freebase远程监督生成
- 53个关系类型
- 训练集：281,270个实体对，18,252个关系事实
- 测试集：96,678个实体对，1,950个关系事实

- 基于CNN的单语句编码器：

  CNN+max, PCNN, CNN+word_att, ResCNN+max

- 基于双向LSTM的单语句编码器：

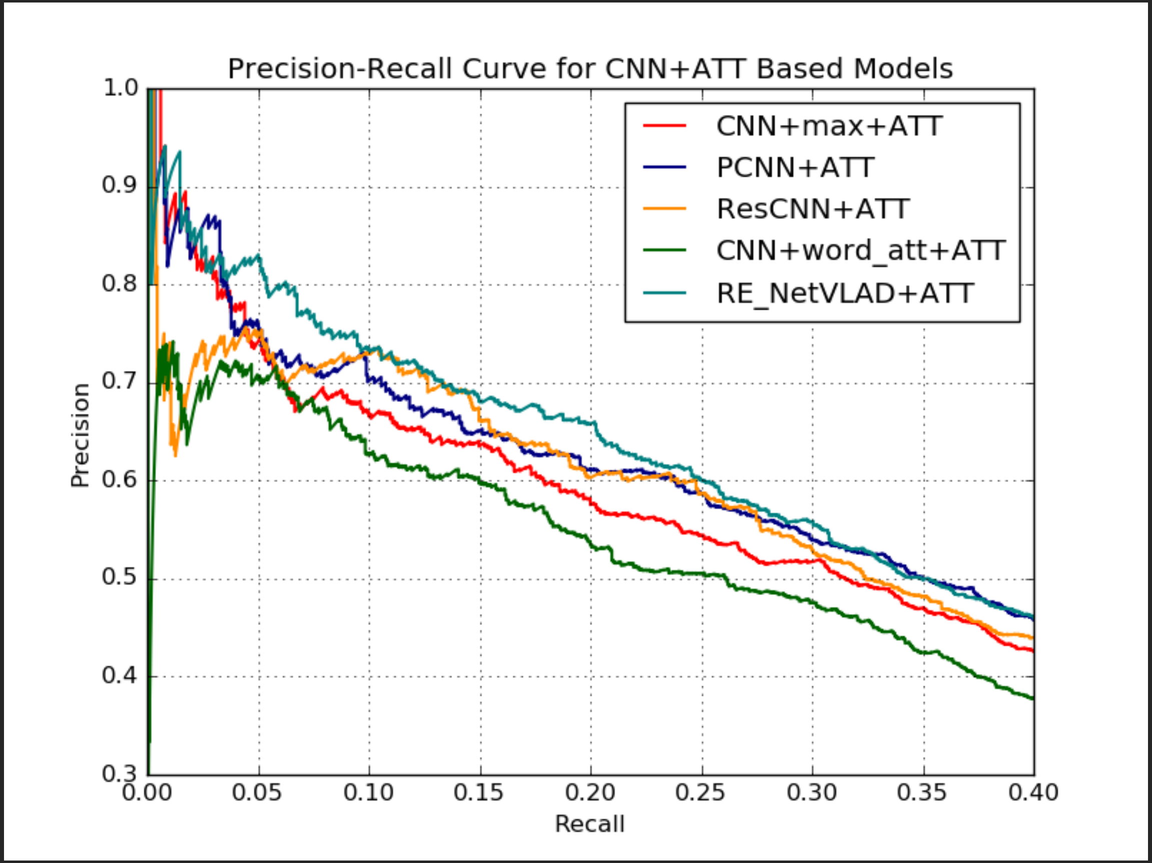
  Bi-LSTM, Bi-LSTM+max, Bi-LSTM+word_att

RE_NetVLAD与基线模型效果对比

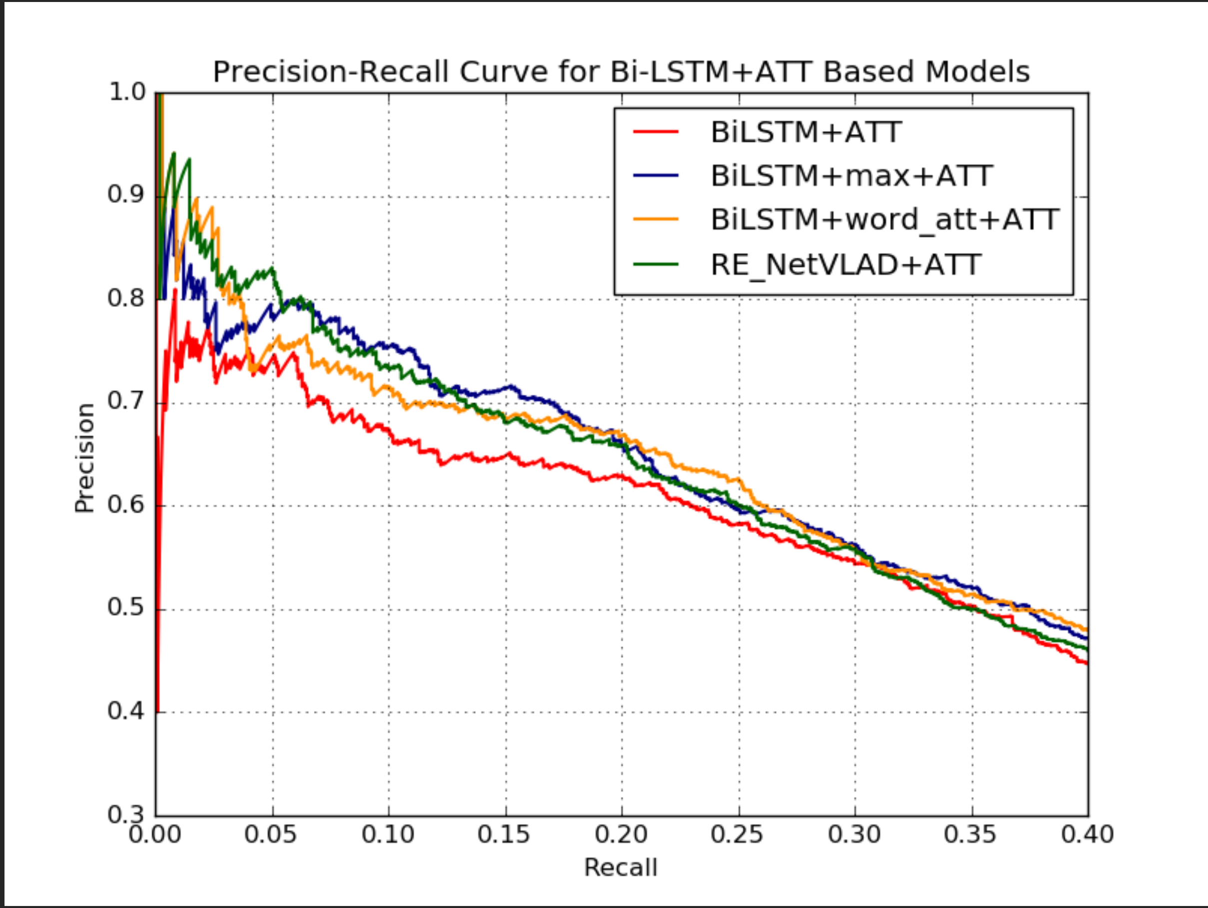
表 4.2　RE_NetVLAD 与基线模型的效果比较。

多语句编码器	AVG					ATT				
P@N(%)	100	200	300	Mean	GAP	100	200	300	Mean	GAP
CNN+max	81.0	71.0	67.0	73.0	35.3	78.0	67.5	66.7	70.7	36.1
PCNN	**82.0**	**76.0**	67.3	**75.1**	36.5	76.0	71.5	70.0	72.5	38.2
CNN+word_att	66.0	64.5	64.7	65.1	33.2	71.0	67.0	64.0	67.3	31.2
ResCNN+max	68.0	65.5	63.0	65.5	34.0	74.0	71.0	**72.3**	72.4	35.2
Bi-LSTM	68.0	69.0	66.7	67.9	35.2	74.0	70.5	66.3	70.3	36.2
Bi-LSTM+max	78.0	71.0	67.7	72.2	**37.4**	77.0	**77.0**	73.7	75.9	39.8
Bi-LSTM+word_att	80.0	73.0	**68.7**	73.9	37.3	76.0	73.0	69.7	72.9	39.5
RE_NetVLAD	69.0	70.0	66.3	68.4	37.2	**82.0**	76.0	**72.3**	**76.8**	**40.0**

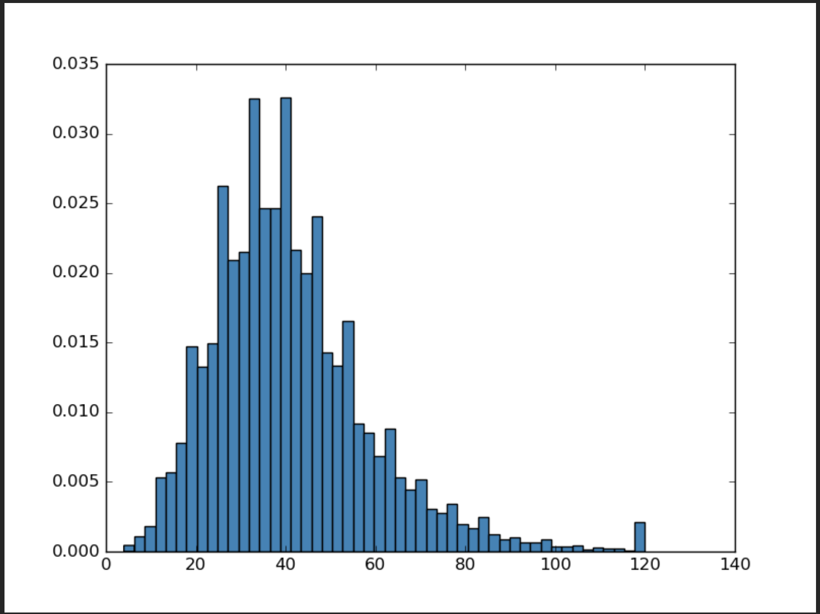
RE_NetVLAD与CNN对比

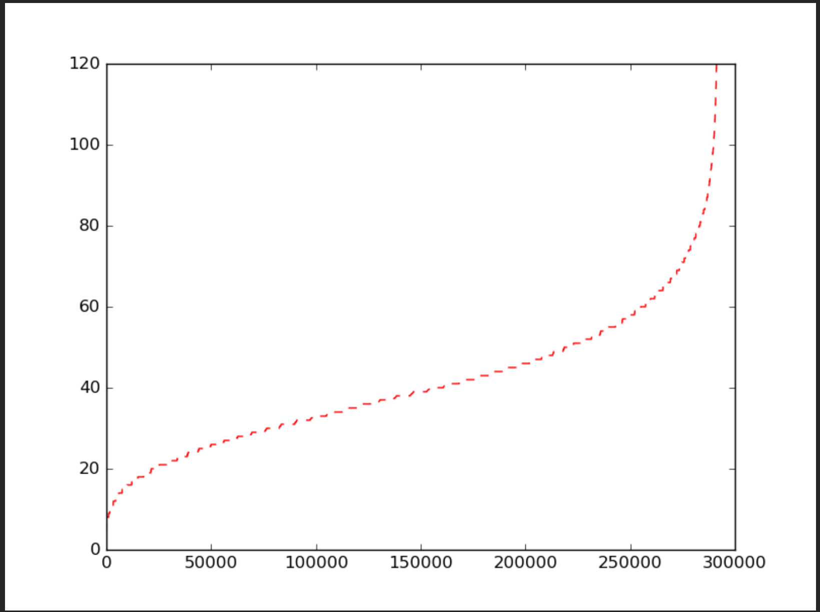


RE_NetVLAD与Bi-LSTM对比

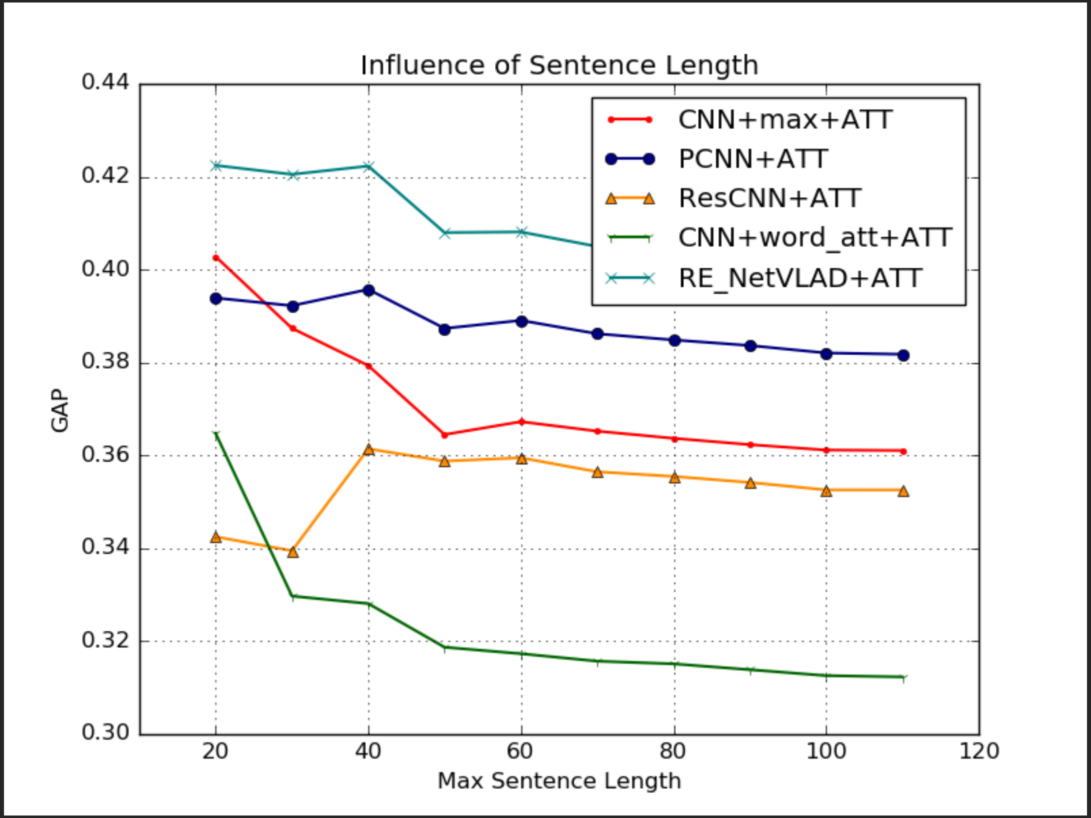


RE_NetVLAD与单语句长度

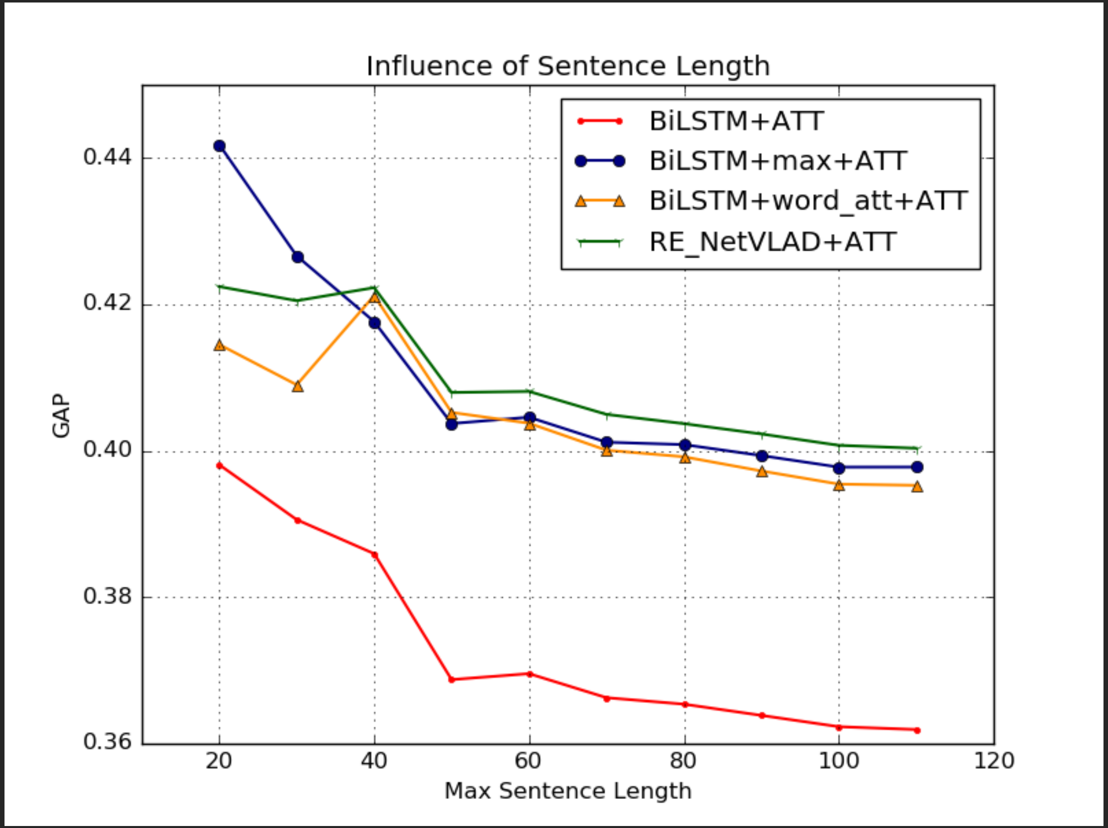




RE_NetVLAD与单语句长度



RE_NetVLAD与单语句长度

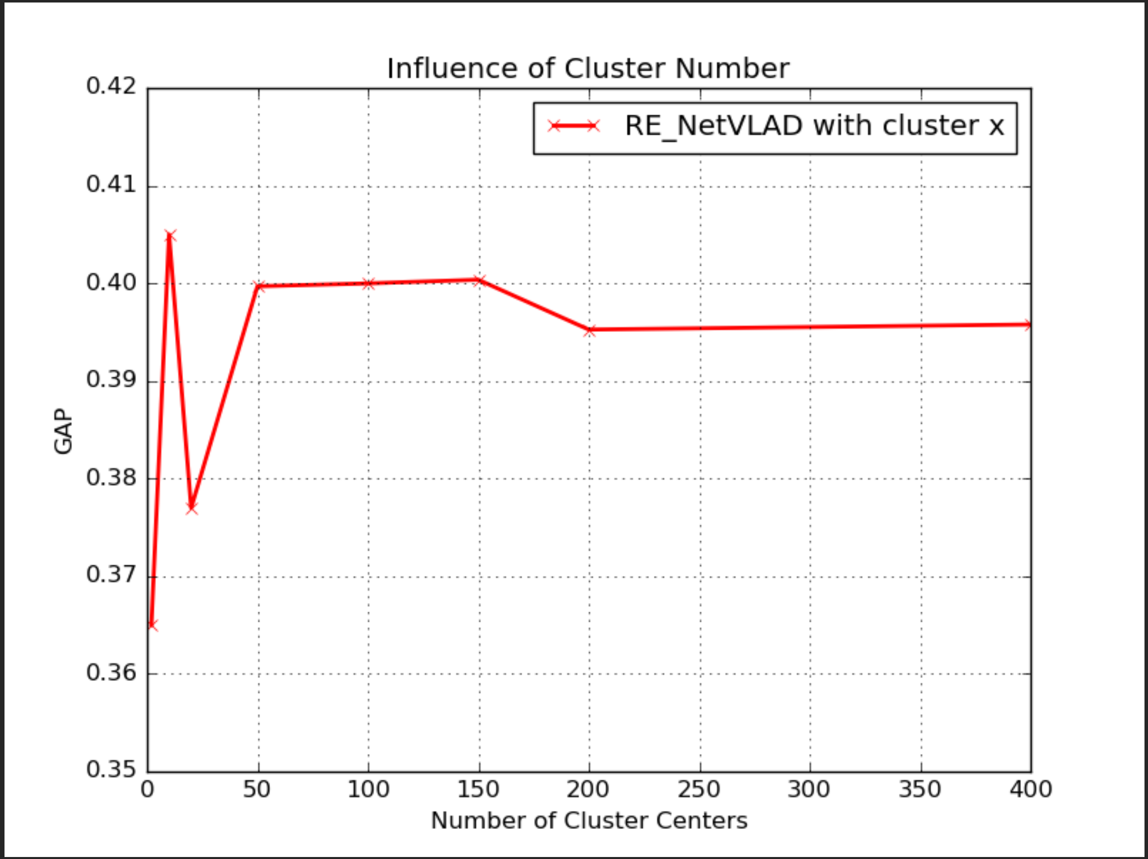


RE_NetVLAD与模型超参数

表 4.3　聚类中心数目 *K* 的大小对于 RE_NetVLAD 效果的影响。

多语句编码器	ATT				
P@N(%)	100	200	300	Mean	GAP
cluster 2	80.0	76.5	74.0	76.8	36.5
cluster 10	82.0	78.0	74.0	**78.0**	**40.5**
cluster 20	73.0	69.5	67.7	70.1	37.7
cluster 50	80.0	76.5	72.7	76.4	40.0
cluster 100	82.0	76.0	72.3	76.8	40.0
cluster 150	82.0	75.5	72.0	76.5	40.0
cluster 200	**85.0**	**78.5**	73.0	78.8	39.5
cluster 400	80.0	73.0	71.7	74.9	39.5

RE_NetVLAD与模型超参数



RE_NetVLAD的语义学分析

表 4.4　语句中不同单词对应的聚类中心的分布。

关系类型	语句中聚类分布
/location/contains	after studying computer science *at* (0.12) the **university_of_pennsylvania in philadelpha**, yang took a *job* (0.14) as a *software* (0.1) engineer at an *investment* (0.37) firm downtown ...
/person/children	he UNK got a great *father* (0.17), barber said, referring *to* (0.1) **archie_manning**, whose UNK career as an UNK quarterback ...
/company/place_founded	the president *of* (0.15) **universal_studios**, who *came* (0.2) *from* (0.38) **los_angeles** ...
/location/neighborhood	...from **carroll_gardens** to the west, is being touted as the next **brooklyn** *neighborhood*(0.48) ...

RE_NetVLAD的语义学分析

表 4.5　不同的聚类中心对应的单词及其权重。

聚类中心 ID	对应的单词及其权重
3	sri_lanka (0.95), west_virginia (0.95), belgium (0.94), italy (0.94), scotland (0.93), texas (0.93) ...
28	karl_lagerfeld (0.95), sumner_redstone (0.95), levi_strauss (0.95), medgar_evers (0.94), william_eggleston (0.94) ...
5	mother (0.73), doctorate(0.68), childhood (0.59), graduate (0.57), child (0.55), doctor (0.52), adolescent (0.48) ...
47	christian (0.95), sundays (0.92), muslim (0.85), islamic (0.83), jihad (0.80), sermons (0.78), hasidic (0.73), jesus (0.68) ...
38	son (0.75), principal (0.74), founder (0.63), bridegroom (0.61), partner (0.60), billionaire (0.58), daughter (0.46), dean (0.44) ...

Part 5 结论与展望

- RE_NetVLAD单模型效果最好
- 在不同设定下都能从效果和效率给出有竞争力的结果
- 对于语句长度和超参数有很强的鲁棒性
- 聚类操作具有语言学意义

- 建立NRE开源框架
- 探索RE_NetVLAD聚类的语言学意义
- 将RE_NetVLAD与CNN LSTM相结合

The End Thanks !